

Kiran Mohan

Assignment 2

Internship Assignment

Cyber Security and Digital Forensics

Assignment 2: Networking Fundamentals

1. Introductory Networking

Introductory Networking

An introduction to networking theory and basic networking tools

Easy

20 min

Start AttackBox

Help

Save Room

16316

Options

Room progress (2%)

Task 1 Introduction

The aim of this room is to provide a beginner's introduction to the basic principles of networking. Networking is a *massive* topic, so this really will just be a brief overview; however, it will hopefully give you some foundational knowledge of the topic, which you can build upon for yourself.

The topics that we're going to cover in this room are:

- The OSI Model
- The TCP/IP Model
- How these models look in practice
- An introduction to basic networking tools

Answer the questions below

Let's get started!

No answer needed

✓ Correct Answer

6

✓ Correct Answer

Which layer tracks communications between the host and receiving computers? Answer with the number of the layer: e.g. if the answer would be "the application layer", then you would enter "7".

5

✓ Correct Answer

Which layer accepts communication requests from applications? Answer with the number of the layer: e.g. if the answer would be "the application layer", then you would enter "7".

7

✓ Correct Answer

Which layer handles logical addressing? Answer with the number of the layer: e.g. if the answer would be "the application layer", then you would enter "7".

3

✓ Correct Answer

When sending data over TCP, what would you call the "bite-sized" pieces of data? Answer with the number of the layer: e.g. if the answer would be "the application layer", then you would enter "7".

Segments

✓ Correct Answer

[Research] Which layer would the FTP protocol communicate with? Answer with the number of the layer: e.g. if the answer would be "the application layer", then you would enter "7".

7

✓ Correct Answer

Hint

Which transport layer protocol would be best suited to transmit a live video? Answer with the number of the layer: e.g. if the answer would be "the application layer", then you would enter "7".

UDP

✓ Correct Answer

will be understood -- regardless of whether they are from the same manufacturer; use the same operating system; or any other

✓ Woop woop! Your answer is correct

Answer the questions below

How would you refer to data at layer 2 of the encapsulation process (with the OSI model)?

Frames

✓ Correct Answer

How would you refer to data at layer 4 of the encapsulation process (with the OSI model), if the UDP protocol has been selected?

Datagrams

✓ Correct Answer

What process would a computer perform on a received message?

De-encapsulation

✓ Correct Answer

Which is the only layer of the OSI model to add a *trailer* during encapsulation?

Data Link

✓ Correct Answer

Does encapsulation provide an extra layer of security (**Aye/Nay**)?

aye

✓ Correct Answer

Task 4 ○ The TCP/IP Model

Task 5 ○ Networking Tools Ping

The Network Interface layer of the TCP/IP model covers the functionality of two layers in the OSI model. These layers are Data Link, and?

Physical

✓ Woop woop! Your answer is correct

Which layer of the TCP/IP model handles the functionality of the OSI network layer?

Internet

✓ Correct Answer

What kind of protocol is TCP?

Connection-based

✓ Correct Answer

Hint

What is SYN short for?

Synchronise

✓ Correct Answer

Hint

What is the second step of the three way handshake?

SYN/ACK

✓ Correct Answer

What is the short name for the "Acknowledgement" segment in the three-way handshake?

ACK

✓ Correct Answer

Task 5 ○ Networking Tools Ping

Task 6 ○ Networking Tools Traceroute

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tryhackme.com

TryHackMe | Introductory Networking

When was the facebook.com domain first registered (Format: DD/MM/YYYY)?

29/03/1997

✓ Correct Answer

✓ Woop woop! Your answer is correct

Perform a whois search on microsoft.com

(Note: Please ensure you have read the task above before attempting the next questions.)

No answer needed

✓ Correct Answer

Which city is the registrant based in?

Redmond

✓ Correct Answer

[OSINT] What is the name of the golf course that is near the registrant address for microsoft.com?

Bellevue Golf Course

✓ Correct Answer

What is the registered Tech Email for microsoft.com?

msnhst@microsoft.com

✓ Correct Answer

Task 8 Networking Tools Dig

Task 9 Further Reading

How likely are you to recommend this room to others?

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tryhackme.com

TryHackMe | Introductory Networking

Domain Name System

What is the first type of DNS server your computer would query when you search for a domain?

Recursive

✓ Correct Answer

✓ Woop woop! Your answer is correct

What type of DNS server contains records specific to domain extensions (i.e. .com, .co.uk*, etc)*? Use the long version of the name.

Top-Level Domain

✓ Correct Answer

Where is the very first place your computer would look to find the IP address of a domain?

Hosts File

✓ Correct Answer

Hint

[Research] Google runs two public DNS servers. One of them can be queried with the IP 8.8.8.8, what is the IP address of the other one?

8.8.4.4

✓ Correct Answer

If a DNS query has a TTL of 24 hours, what number would the dig query show?

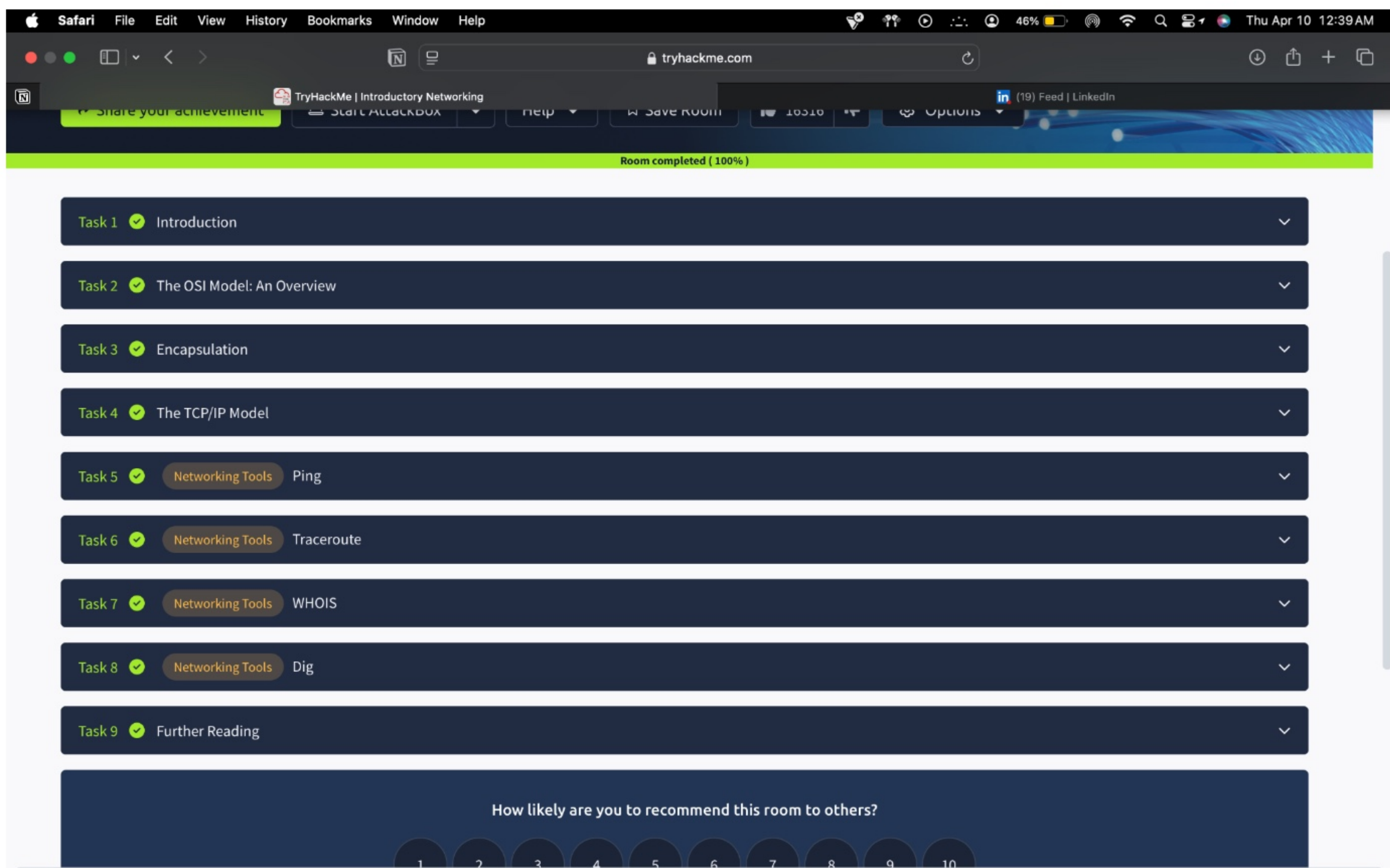
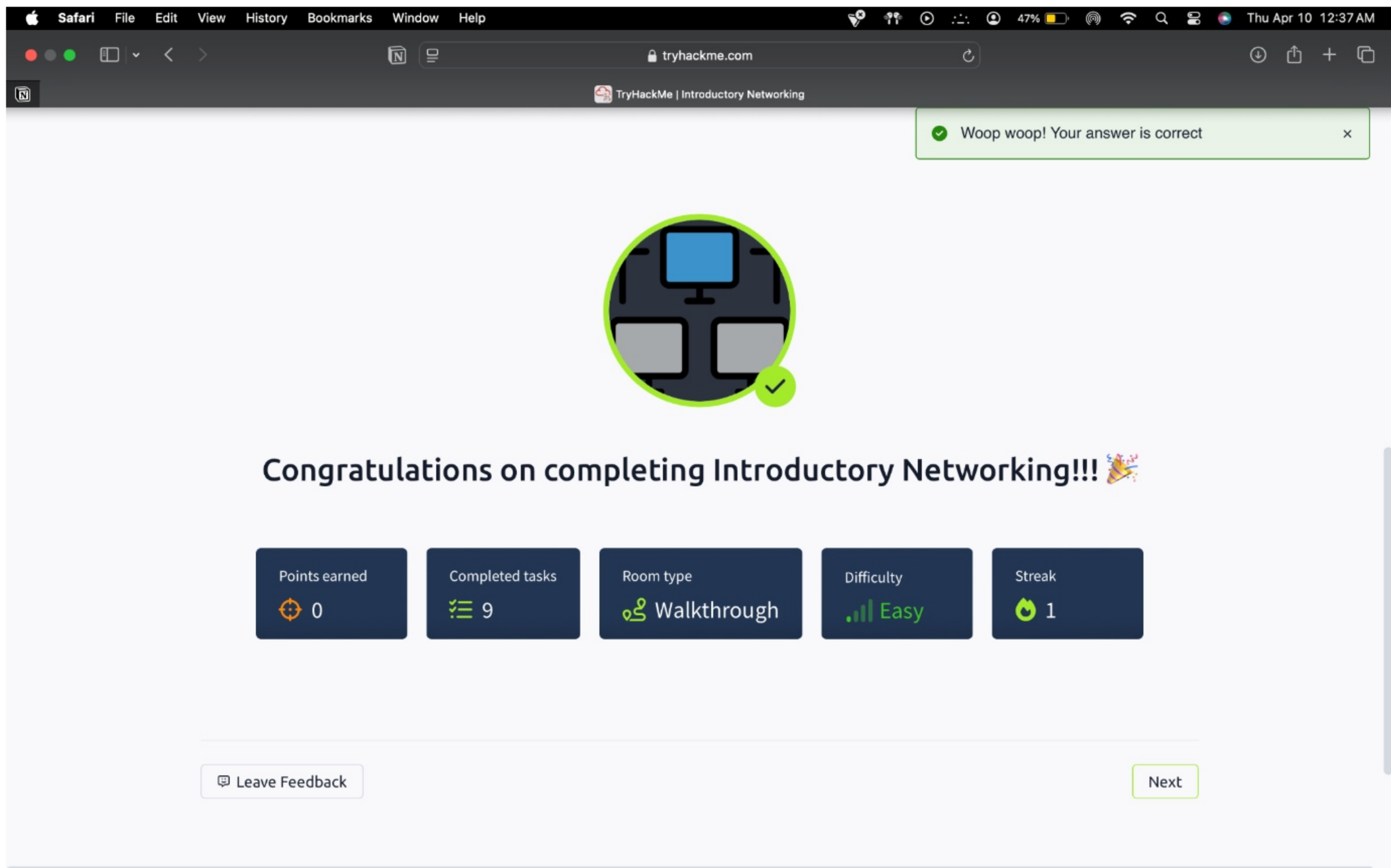
86400

✓ Correct Answer

Task 9 Further Reading

How likely are you to recommend this room to others?

12345678910



2. What is Networking?

Task 1 What is Networking?

Task 2 What is the Internet?

Task 3 Identifying Devices on a Network

Task 4 Ping (ICMP)

Task 5 Continue Your Learning: Intro to LAN

How likely are you to recommend this room to others?

1

2

3

4

5

6

7

8

9

10

Submit now

Created by

Room Type

Users in Room

Created

tryhackme cmnatic

Free Room. Anyone can deploy virtual machines in the room (without being subscribed)!

755,371

1379 days ago

```
user@thm:~$ ping -c 4 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data
64 bytes from 8.8.8.8: icmp_seq=1 ttl=56 time=9.95 ms
64 bytes from 8.8.8.8: icmp_seq=1 ttl=56 time=8.38 ms
64 bytes from 8.8.8.8: icmp_seq=1 ttl=56 time=9.77 ms
64 bytes from 8.8.8.8: icmp_seq=1 ttl=56 time=7.28 ms
--- 8.8.8.8 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time
4006ms
rtt min/avg/max/mdev = 8.132/9.428/10.957/1.057 ms
Flag: THM{I_PINGED_THE_SERVER}
user@thm:~$ ping -c 4
```

TryHackMe | What is Networking?

(19) Kiran Mohan | LinkedIn



Networks come in all shapes and sizes, which is something that we will also come on to discuss throughout this module.

Answer the questions below

What is the key term for devices that are connected together?

☐ Group

☐ Collection

☒ Network

☐ System

✓ Correct Answer

Task 2 ☐ What is the Internet?

tryhackme.com

TryHackMe | What is Networking?

(19) Licenses & certifications | Kiran Mohan | LinkedIn

Woop woop! Your answer is correct

As previously stated, the Internet is made up of many small networks all joined together. These small networks are called private networks, where networks connecting these small networks are called public networks -- or the Internet! So, to recap, a network can be one of two types:

- A private network
- A public network

Devices will use a set of labels to identify themselves on a network, which we will come onto in the task below.

Answer the questions below

Who invented the World Wide Web?

☐ Larry Page

☐ Marc Andreessen

☐ Vint Cerf

☒ Tim Berners-Lee

✓ Correct Answer

🔍 Hint

Task 3

Identifying Devices on a Network

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Woop woop! Your answer is correct

6

8

✓ Correct Answer

🔍 Hint

What does the term "MAC" stand for?

☐ Machine Access Control

☐ Main Access Channel

☒ Media Access Control

☐ Managed Access Control

✓ Correct Answer

Deploy the interactive lab using the "View Site" button and spoof your MAC address to access the site. What is the flag?

THM{YOU_GOT_ON_TRYHACKME}

✓ Correct Answer

Task 4

Ping (ICMP)

Task 5

Continue Your Learning: Intro to LAN

THM{YOU_GOT_ON_TRYHACKME}

Bob

00:12:32:2F:33:39

Alice

00:12:32:2F:33:39

Request Site

Free Hotel Wifi

Answer the questions below

What protocol does ping use?

☒ ICMP

☐ UDP

☐ HTTP

☐ TCP

✓ Correct Answer

What is the syntax to ping 10.10.10.10?

ping 10.10.10.10

✓ Correct Answer

What flag do you get when you ping 8.8.8.8?

THM{I_PINGED_THE_SERVER}

✓ Correct Answer

Task 5

Continue Your Learning: Intro to LAN

How likely are you to recommend this room to others?

1

2

3

4

5

6

7

8

9

10

✓ Woop woop! Your answer is correct

```
user@thm:~$ ping -c 4 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data
64 bytes from 8.8.8.8: icmp_seq=1 ttl=56 time=9.95 ms
64 bytes from 8.8.8.8: icmp_seq=1 ttl=56 time=8.38 ms
64 bytes from 8.8.8.8: icmp_seq=1 ttl=56 time=9.77 ms
64 bytes from 8.8.8.8: icmp_seq=1 ttl=56 time=7.28 ms
--- 8.8.8.8 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time
4006ms
rtt min/avg/max/mdev = 8.132/9.428/10.957/1.057 ms
Flag: THM{I_PINGED_THE_SERVER}
user@thm:~$ ping -c 4
```

Free Hotel Wifi

Network Ping

Congratulations on completing What is Networking?!!! 🎉

Points earned
80

Completed tasks
5

Room type
Walkthrough

Difficulty
Info

Streak
1

Leave Feedback

Next

3. Introduction to Networking on Hackthebox.

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Thu Apr 10 8 11 PM

Introduction to Networking

Fun Story

During COVID, I was tasked to perform a **Physical Penetration Test** across state lines, and my state was under a **stay at home** order. The company I was testing had minimal staff in the office. I decided to purchase an expensive printer and exploited it to put a reverse shell in it, so when it connected to the network, it would send me a shell (remote access). Then I shipped the printer to the company and sent a phishing email thanking the staff for coming in and explaining that the printer should allow them to print or scan things more quickly if they want to bring some stuff home to WFH for a few days. The printer was hooked up almost instantly, and their domain administrator's computer was kind enough to send the printer his credentials!

If the client had designed a secure network, this attack probably would not have been possible for many reasons:

- Printer should not have been able to talk to the internet
- Workstation should not have been able to communicate to the printer over port 445
- Printer should not be able to initiate connections to workstations. In some cases, printer/scanner combinations should be able to communicate to a mail server to email scanned documents.

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Introduction to Networking

used, which enable a significantly higher data throughput than the Internet. The transmission speed between two remote nodes is comparable to communication within a **LAN**.

Internationally operating network operators provide the infrastructure for **MANs**. Cities wired as **Metropolitan Area Networks** can be integrated supra-regionally in **Wide Area Networks (WAN)** and internationally in **Global Area Networks (GAN)**.

PAN / WPAN

Modern end devices such as smartphones, tablets, laptops, or desktop computers can be connected ad hoc to form a network to enable data exchange. This can be done by cable in the form of a **Personal Area Network (PAN)**.

The wireless variant **Wireless Personal Area Network (WPAN)** is based on Bluetooth or Wireless USB technologies. A **wireless personal area network** that is established via Bluetooth is called **Piconet**. **PANs** and **WPANs** usually extend only a few meters and are therefore not suitable for connecting devices in separate rooms or even buildings.

In the context of the **Internet of Things (IoT)**, **WPANs** are used to communicate control and monitor applications with low data rates. Protocols such as Insteon, Z-Wave, and ZigBee were explicitly designed for smart homes and home automation.

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Introduction to Networking

component via its previous nodes to the computer system.

Daisy Chain Topology

```
graph LR; A[Host A] --- B[Host B]; B --- C[Host C]; C --- D[Host D]; D --- E[Host E]
```

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```
graph LR; B[Host B] --- Internet[Internet]; B -- HTTP Request --> Internet
```

(Non-) Transparent Proxy

All these proxy services act either **transparently** or **non-transparently**.

With a **transparent proxy**, the client doesn't know about its existence. The transparent proxy intercepts the client's communication requests to the Internet and acts as a substitute instance. To the outside, the transparent proxy, like the non-transparent proxy, acts as a communication partner.

If it is a **non-transparent proxy**, we must be informed about its existence. For this purpose, we and the software we want to use are given a special proxy configuration that ensures that traffic to the Internet is first addressed to the proxy. If this configuration does not exist, we cannot communicate via the proxy. However, since the proxy usually provides the only communication path to other networks, communication to the Internet is generally cut off without a corresponding proxy configuration.

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is transmitted to the receiver. The receiver reverses the process and unpacks the data on each layer with the header information. After that, the application finally uses the data. This process continues until all data has been sent and received.

Packet Transfer

The diagram illustrates the encapsulation process for packet transfer over an Ethernet network. It shows a 'Sender' and a 'Receiver' connected via an 'Ethernet' network. On the sender side, data is encapsulated through several layers: Data (with header H), TCP (with header H), IP (with header H), and MAC (with header H). The resulting frame is then converted into bits (10010010001011...) for transmission. On the receiver side, the process is reversed: bits are received, converted back into a frame, and then the data is extracted from the various headers (MAC, IP, TCP, Data) to be used by the application. Labels in the diagram include: Data, H, TCP, IP, MAC, Sequence / Datagram, Packet, Frame, Bits, and Ethernet.

For us, as penetration testers, both reference models are useful. With **TCP/IP**, we can quickly understand how the entire connection is established, and with **ISO**, we can take it apart piece by piece and analyze it in detail. This often happens when we can listen to and intercept specific network traffic. We then have to analyze this traffic accordingly, going into more detail in the **Network Traffic Analysis** module. Therefore, we should familiarize ourselves with both reference models and understand and internalize them in the best possible way.

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described. Each layer offers services for use to the layer directly above it. To make these services available, the layer uses the services of the layer below it and performs the tasks of its layer.

If two systems communicate, all seven layers of the **OSI** model are run through at least **twice**, since both the sender and the receiver must take the layer model into account. Therefore, a large number of different tasks must be performed in the individual layers to ensure the communication's security, reliability, and performance.

When an application sends a packet to the other system, the system works the layers shown above from layer **7** down to layer **1**, and the receiving system unpacks the received packet from layer **1** up to layer **7**.

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wireless networks

Virtual Private Networks

Vendor Specific Information

Connection Establishment

Key Exchange Mechanisms

Authentication Protocols

TCP/UDP Connections

Cryptography

My Workstation

OFFLINE

Start Instance

1 / 1 spawns left

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Introduction to Networking

Me

1. Layer PhysicalNetwork Card

I. LINK

Key Exchange Mechanisms

Authentication Protocols

TCP/UDP Connections

Cryptography

The most important tasks of TCP/IP are:

Task	Protocol	Description
Logical Addressing	IP	Due to many hosts in different networks, there is a need to structure the network topology and logical addressing. Within TCP/IP, IP takes over the logical addressing of networks and nodes. Data packets only reach the network where they are supposed to be. The methods to do so are network classes, subnetting, and CIDR.
Routing	IP	For each data packet, the next node is determined in each node on the way from the sender to the receiver. This way, a data packet is routed to its receiver, even if its location is unknown to the sender.
Error & Control Flow	TCP	The sender and receiver are frequently in touch with each other via a virtual connection. Therefore control messages are sent continuously to check if the connection is still established.
Application Support	TCP	TCP and UDP ports form a software abstraction to distinguish specific applications and their communication links.
Name Resolution	DNS	DNS provides name resolution through Fully Qualified Domain Names (FQDN) in IP addresses, enabling us to reach the desired host with the specified name on the internet.

My Workstation

OFFLINE

Start Instance

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between the network nodes. Since direct communication between the sender and the receiver is not always possible due to the different subnets, packets must be forwarded from nodes (routers) that are on the way. Forwarded packets do not reach the higher layers but are assigned a new intermediate destination and sent to the next node.

wireless networks

Virtual Private Networks

Vendor Specific Information

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assignment between IPv4 address and network classes (A, B, C, D, E). The division is based on the subnet mask or the so-called **CIDR suffix**, which allows the bitwise division of the IPv4 address space and thus into **subnets** of any size. The **CIDR suffix** indicates how many bits from the beginning of the IPv4 address belong to the network. It is a notation that represents the **subnet mask** by specifying the number of **1-bits** in the subnet mask.

Let us stick to the following IPv4 address and subnet mask as an example:

- IPv4 Address: **192.168.10.39**
- Subnet mask: **255.255.255.0**

Now the whole representation of the IPv4 address and the subnet mask would look like this:

- CIDR: **192.168.10.39/24**

The CIDR suffix is, therefore, the sum of all ones in the subnet mask.

IP Addresses

Octet:	1st	2nd	3rd	4th
Binary:	1111 1111	. 1111 1111	. 1111 1111	. 0000 0000 (/24)
Decimal:	255	. 255	. 255	. 0

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+ 2 Submit the broadcast address of the following CIDR: 10.200.20.0/27

10.200.20.31

Submit

+ 3 Split the network 10.200.20.0/27 into 4 subnets and submit the network address of the 3rd subnet as the answer.

10.200.20.16

Submit

+ 3 Split the network 10.200.20.0/27 into 4 subnets and submit the broadcast address of the 2nd subnet as the answer.

10.200.20.15

Submit

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+10 Streak pts

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done using tools like [Ettercap](#) or [Cain & Abel](#) in which we send falsified ARP messages over a LAN. The goal is to associate our MAC address with the IP address of a legitimate device on the company's network, effectively allowing us to intercept traffic intended for the legitimate device. For example, this could look like the following:

MAC Addresses

```
1  0.000000 10.129.12.100 -> 10.129.12.101 ARP 60 10.129.12.101 is at AA:AA:AA:AA:AA
2  0.000015 10.129.12.100 -> 10.129.12.255 ARP 60 Who has 10.129.12.101? Tell 10.12
3  0.000030 10.129.12.101 -> 10.129.12.100 ARP 60 10.129.12.101 is at BB:BB:BB:BB:BB
4  0.000045 10.129.12.100 -> 10.129.12.101 ARP 60 10.129.12.101 is at AA:AA:AA:AA:AA
```

The first and fourth lines show us (10.129.12.100) sending falsified ARP messages to the target, associating its MAC address with its IP address (10.129.12.101). The second and third lines show the target sending an ARP request and replying to our MAC address. This indicates that we have poisoned the target's ARP cache and that all traffic intended for the target will now be sent to our MAC address.

We can use ARP poisoning to perform various activities, such as stealing sensitive information, redirecting traffic, or launching MITM attacks. However, to protect against ARP spoofing, it is important to use secure network protocols, such as IPSec or SSL, and to implement security measures, such as firewalls and intrusion detection systems.

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- Full IPv6: `fe80:0000:0000:0000:dd80:b1a9:6687:2d3b/64`
- Short IPv6: `fe80::dd80:b1a9:6687:2d3b/64`

An IPv6 address consists of two parts:

- **Network Prefix** (network part)
- **Interface Identifier** also called **Suffix** (host part)

The **Network Prefix** identifies the network, subnet, or address range. The **Interface Identifier** is formed from the **48-bit MAC** address (which we will discuss later) of the interface and is converted to a **64-bit address** in the process. The default prefix length is **/64**. However, other typical prefixes are **/32**, **/48**, and **/56**. If we want to use our networks, we get a shorter prefix (e.g. **/56**) than **/64** from our provider.

In RFC 5952, the aforementioned IPv6 address notation was defined:

- All alphabetical characters are always written in lower case.
- All leading zeros of a block are always omitted.
- One or more consecutive blocks of **4 zeros** (hex) are shortened by two colons (::).
- The shortening to two colons (::) may only be performed **once** starting from the left.

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Introduction to Networking

Feed

Asynchronous JavaScript and XMLAJAXWeb development technique that allows creating dynamic web pages using JavaScript and XML/JSON.

Internet Server Application Programming InterfaceISAPIAllows to create performance-oriented web extensions for web servers using a set of APIs.

Uniform Resource IdentifierURIIt is a syntax used to identify a resource on the Internet.

Uniform Resource LocatorURLSubset of URI that identifies a web page or another resource on the Internet, including the protocol and the domain name.

Internet Key ExchangeIKEIKE is a protocol used to set up a secure connection between two computers. It is used in virtual private networks (VPNs) to provide authentication and encryption for data transmission, protecting the data from outside eavesdropping and tampering.

Generic Routing EncapsulationGREThis protocol is used to encapsulate the data being transmitted within the VPN tunnel.

Remote ShellRSHIt is a program under Unix that allows executing commands and programs on a remote computer.

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Introduction to Networking

CANCEL

REGISTER

OPTIONS

Requests information about the capabilities of a SIP server or user agent, such as the types of media it supports.

Information Disclosure

However, SIP allows us to enumerate existing users for potential attacks. This can be done for various purposes, such as determining a user's availability, finding out information about the user's capabilities or services, or performing brute-force attacks on user accounts later on.

One of the possible ways to enumerate users is the SIP **OPTIONS** request. It is a method used to request information about the capabilities of a SIP server or user agents, such as the types of media it supports, the codecs it can decode, and other details. The **OPTIONS** request can probe a SIP server or user agent for information or test its connectivity and availability.

During our analysis, it is possible to discover a **SEPxxxx.cnf** file, where **xxxx** is a unique identifier, is a configuration file used by Cisco Unified Communications Manager, formerly known as Cisco CallManager, to define the settings and parameters for a Cisco Unified IP Phone. The file specifies the phone model, firmware version, network settings, and other details.

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Introduction to Networking

1. WPA-Personal

2. WPA-Enterprise

WPA-Personal, designed for home and small business networks, and WPA-Enterprise, designed for larger organizations and uses a centralized authentication server (e.g., RADIUS or TACACS+) to verify the identity of clients.

MAC Filtering

MAC filtering is a security measure that allows a WAP to accept or reject connections from specific devices based on their MAC addresses. By configuring the WAP to accept connections only from devices with approved MAC addresses, it is possible to prevent unauthorized devices from connecting to the network.

Deploying EAP-TLS

EAP-TLS is a security protocol used to authenticate and encrypt wireless communications. It uses digital certificates and PKI to verify the identity of clients and establish secure connections. Deploying EAP-TLS can help to harden a WAP by providing strong authentication and encryption for wireless communications, which can protect against unauthorized access to the network and the interception of sensitive data.

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Introduction to Networking

Encapsulating Security Payload (ESP)

UDP/4500

ESP is also a protocol that provides encryption and authentication for IP datagrams. It is used to encrypt the VPN traffic between the VPN client and the VPN server, using the keys that were negotiated with IKE.

These protocols are necessary for facilitating IPsec VPN traffic because they provide the security and encryption that are required for secure communication over the public internet. Without these protocols, the VPN traffic would be vulnerable to interception and tampering.

PPTP

Point-to-Point Tunneling Protocol (PPTP) is a network protocol that enables the creation of VPNs by establishing a secure tunnel between the VPN client and server, encapsulating the data transmitted within this tunnel. Originally an extension of the Point-to-Point Protocol (PPP), PPTP is supported by many operating systems.

However, due to its known vulnerabilities, PPTP is no longer considered secure. It can tunnel protocols such as IP, IPX, or NetBEUI via IP, but has been largely replaced by more secure VPN protocols like L2TP/IPsec, IPsec/IKEv2, and OpenVPN. Since 2012, the use of PPTP has declined because its authentication method, MSCHAPv2, employs the outdated DES encryption, which can be easily cracked with specialized hardware.

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The shown message contains information about the device itself, such as the device name, IP address, port name, and functionality of the router, as well as information about the operating system and hardware platform of the device. Besides, we can see in the first line from the **CDPv2** that we are dealing with the **Cisco Discovery Protocol**.

For comparison, we can look at another protocol called Spanning Tree Protocol (**STP**). The **STP** is a network protocol that ensures no loops in a network with multiple connections between switches. There are no loops, and it prevents data packets from circulating in a loop and congesting the network.

STP Network Traffic

VLANs

22:14:11.563654 STP 802.1w, Rapid STP, Flags [Learn, Forward], bridge-id 8001.00:11:22
root-id 8001.AA:AA:AA:AA:AA, cost 0, port-id 8001, message-age 0.00s, max-a

In this example, we see that an **STP** message was sent containing information about the root switch, the MAC address of the root switch, the ID of the port over which the message was sent, and other configuration parameters such as the maximum aging time, hello time, and forward delay.

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Aggressive mode is an alternative mode for **IKE** that provides **faster performance** by reducing the number of round trips and message exchanges required for key exchange. In this mode, the key exchange process is performed in **two phases**, with all security parameters and keys being exchanged in the first phase. However, this can provide faster performance but may also reduce the security of the IKE session compared to the main mode since the **aggressive mode** does not provide identity protection.

Pre-Shared Keys

In IKE, a **Pre-Shared Key (PSK)** is a secret value shared between the two parties involved in the key exchange. This key is used to authenticate the parties and establish a shared secret that encrypts subsequent communication. The use of a PSK is optional in IKE, and whether or not to use one depends on the specific requirements and constraints of the situation. However, if a **PSK** is used, it must be securely exchanged between the two parties before the key exchange process begins. This can be done through a secure out-of-band channel, such as a separate communication channel, or by physically exchanging the key.

The main advantage of using a PSK is that it provides an additional layer of security by allowing the parties to authenticate with each other. However, using a PSK also has some limitations and drawbacks. For example, it can be difficult to exchange the key securely, and if the key is compromised through a MITM attack, the security of the IKE session may be compromised.

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For example, LEAP and PEAP may be used to authenticate wireless clients when they access the wireless network or to authenticate remote employees connecting to the network via a VPN. In these cases, using SSH or HTTPS would be challenging to implement, as they are designed for different purposes. In terms of security, **PEAP** is generally more secure than LEAP because it uses a server-side public key certificate to authenticate the server and encrypting **MSCHAPv2** hash, while **LEAP** relies on a shared secret that is negotiated between the client and the server and does **not** encrypt **MSCHAPv2** hashes. PEAP also supports the use of stronger encryption algorithms, such as AES and 3DES, for encrypting the authentication information, whereas LEAP uses the weaker RC4 algorithm.

However, both protocols are vulnerable to specific attacks and have been largely replaced by more secure protocols such as EAP-TLS.

As physical connections, protocols with **SSL** or **TLS** are used by default, such as **SSH** or **HTTPS**. Both protocols use robust encryption algorithms to protect the authentication information transmitted between the client and the server. This helps to prevent interception or to tamper the authentication data, which is essential for maintaining the security of the authentication process. Also, they support using digital certificates and **PKI** for authenticating the server to the client. This ensures that the client can verify the server's identity and helps to prevent MITM attacks. SSH and HTTPS are widely used and well-supported protocols, with implementations available for various operating systems and devices and makes them easy to use and deploy in a variety of environments.

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data. Instead, the data is sent directly to the target host without any prior connection.

When **traceroute** is used with UDP, we will receive a **Destination Unreachable** and **Port Unreachable** message when the UDP datagram packet reaches the target device. Generally, UDP packets are sent using **traceroute** on Unix hosts.

Blind Spoofing

Blind spoofing, is a method of data manipulation attack in which an attacker sends false information on a network without seeing the actual responses sent back by the target devices. It involves manipulating the IP header field to indicate false source and destination addresses. For example, we send a TCP packet to the target host with false source and destination port numbers and a false **Initial Sequence Number (ISN)**. The **ISN** is a field in the TCP header that is used to specify the sequence number of the first TCP packet in a connection. The ISN is set by the sender of a TCP packet and sent to the receiver in the TCP header of the first packet. This can cause the target host to establish a connection with us without receiving the connection.

This attack is commonly used to disrupt the integrity of network connections or to break connections between network devices. It can also be used to monitor network traffic or to intercept information sent by network devices.

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Mode	Description
Electronic Code Book (ECB) mode	ECB mode is generally not recommended for use due to its susceptibility to certain types of attacks. Furthermore, it does not hide data patterns efficiently. As a result, statistical analysis can reveal elements of clear-text messages, for example, in web applications.
Cipher Block Chaining (CBC) mode	CBC mode is generally used to encrypt messages like disk encryption and e-mail communication. This is the default mode for AES and is also used in software like TrueCrypt, VeraCrypt, TLS, and SSL.
Cipher Feedback (CFB) mode	CFB mode is well suited for real-time encryption of a data stream, e.g., network communication encryption or encryption/decryption of files in transit like Public-Key Cryptography Standards (PKCS) and Microsoft's BitLocker.
Output Feedback (OFB) mode	OFB mode is also used to encrypt a data stream, e.g., to encrypt real-time communication. However, this mode is considered better for the data stream because of how the key stream is generated. We can find this mode in PKCS but also in the SSH protocol.
Counter (CTR) mode	CTR mode encrypts real-time data streams AES uses, e.g., network communication, disk encryption, and other real-time scenarios where data is processed. An example of this would be IPsec or Microsoft's BitLocker.
Galois/Counter (GCM) mode	GCM is used in cases where confidentiality and integrity need to be protected together, such as wireless communications, VPNs, and other secure communication protocols.

Each mode has its characteristics and is more suitable for certain use cases. The choice of encryption mode depends on the application's requirements and the security objectives to be achieved.

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Conclusion

In this module, we cover

- The types and uses of encryption
- Proxies
- Different topologies
- Models and protocols
- OSI & TCP/IP
- IPv4 & IPv6
- Subnetting

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